fn make_rappor

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This proof resides in **"contrib"** because it has not completed the vetting process.

Proves soundness of `make_rappor` in `mod.rs` at commit 304ef9c2 (outdated[1]).
TODO: algorithm outline

1 Hoare Triple

1.1 Preconditions

- Variable

2 Proof

$f/2$ refers to the probability of

Theorem 2.1. [1] `make_rappor` satisfies ε-DP where

$$\varepsilon = 2m \log\left(\frac{2-f}{f}\right) \tag{1}$$

Lemma 2.2.

$$P[y_i = 1 \mid x_i = 1] = 1 - \frac{1}{2}f \tag{2}$$

$$P[y_i = 1 \mid x_i = 0] = \frac{1}{2}f \tag{3}$$

Proof. Let $Y = y_1, ..., y_k$ be a randomised report generated by `make_rappor`. Then the probability of observing any given report Y is $P[Y = y|X = x]$. $x = x_1, ..., x_k$ is a single Boolean vector with at most m ones. Without loss of generality assume that $x^* = \{x_1 = 1, ..., x_m = 1, x_{m+1} = 0, ..., x_k = 0\}$, then we have

$$P[Y = y \mid X = x^*] = \prod_{i=1}^{m} \left(\frac{1}{2}f\right)^{1-y_i} \left(1 - \frac{1}{2}f\right)^{y_i} \times \prod_{i=m+1}^{k} \left(\frac{1}{2}f\right)^{y_i} \left(1 - \frac{1}{2}f\right)^{1-y_i} \tag{4}$$

$$= \left(\frac{1}{2}f\right)^{\sum_{i=1}^{m} 1-y_i} \times \left(1 - \frac{1}{2}f\right)^{\sum_{i=1}^{m} y_i} \times \left(\frac{1}{2}f\right)^{\sum_{i=m+1}^{k} y_i} \times \left(\frac{1}{2}f\right)^{\sum_{i=m+1}^{k} 1-y_i} \tag{5}$$

[1] See new changes with `git diff 304ef9c2..bf1c5f54 rust/src/measurements/rappor/mod.rs`

Then let D be the ratio of two such conditional probabilities with distinct values x_1 and x_2, and let S be the range of `make_rappor`.

$$
\begin{aligned}
D &= \frac{P[Y \in S \mid X = x_1]}{P[Y \in S \mid X = x_2]} && (6)\\
&= \frac{\sum_{y \in S} P[Y = y \mid X = x_1]}{\sum_{y \in S} P[Y = y \mid X = x_2]} && (7)\\
&\leq \max_{y \in S} \frac{P[Y = y \mid X = x_1]}{P[Y = y \mid X = x_2]} && (8)\\
&= \max_{y \in S} \frac{\prod_{i=1}^{m} \left(\frac{1}{2}f\right)^{1-y_i} \left(1 - \frac{1}{2}f\right)^{y_i} \times \prod_{i=m+1}^{k} \left(\frac{1}{2}f\right)^{y_i} \left(1 - \frac{1}{2}f\right)^{1-y_i}}{\prod_{i=1}^{m} \left(\frac{1}{2}f\right)^{y_i} \left(1 - \frac{1}{2}f\right)^{1-y_i} \times \prod_{i=m+1}^{2m} \left(\frac{1}{2}f\right)^{1-y_i} \left(1 - \frac{1}{2}f\right)^{y_i} \times \prod_{i=2m+1}^{k} \left(\frac{1}{2}f\right)^{y_i} \left(1 - \frac{1}{2}f\right)^{1-y_i}} && (9)\\
&= \max_{y \in S} \frac{\prod_{i=1}^{m} \left(\frac{1}{2}f\right)^{1-y_i} \left(1 - \frac{1}{2}f\right)^{y_i} \times \prod_{i=m+1}^{2m} \left(\frac{1}{2}f\right)^{y_i} \left(1 - \frac{1}{2}f\right)^{1-y_i} \times \prod_{i=2m+1}^{k} \left(\frac{1}{2}f\right)^{y_i} \left(1 - \frac{1}{2}f\right)^{1-y_i}}{\prod_{i=1}^{m} \left(\frac{1}{2}f\right)^{y_i} \left(1 - \frac{1}{2}f\right)^{1-y_i} \times \prod_{i=m+1}^{2m} \left(\frac{1}{2}f\right)^{1-y_i} \left(1 - \frac{1}{2}f\right)^{y_i} \times \prod_{i=2m+1}^{k} \left(\frac{1}{2}f\right)^{y_i} \left(1 - \frac{1}{2}f\right)^{1-y_i}} && (10)\\
&= \max_{y \in S} \frac{\prod_{i=1}^{m} \left(\frac{1}{2}f\right)^{1-y_i} \left(1 - \frac{1}{2}f\right)^{y_i} \times \prod_{i=m+1}^{2m} \left(\frac{1}{2}f\right)^{y_i} \left(1 - \frac{1}{2}f\right)^{1-y_i}}{\prod_{i=1}^{m} \left(\frac{1}{2}f\right)^{y_i} \left(1 - \frac{1}{2}f\right)^{1-y_i} \times \prod_{i=m+1}^{2m} \left(\frac{1}{2}f\right)^{1-y_i} \left(1 - \frac{1}{2}f\right)^{y_i}} && (11)\\
&= \max_{y \in S} \left[\prod_{i=1}^{m} \left(\frac{1}{2}f\right)^{2(1-y_i)} \left(1 - \frac{1}{2}f\right)^{2y_i} \times \prod_{i=m+1}^{2m} \left(\frac{1}{2}f\right)^{2y_i} \left(1 - \frac{1}{2}f\right)^{2(1-y_i)}\right] && (12)
\end{aligned}
$$

12 is maximised when $y_1 = 1, ..., y_m = 1$, and $y_{m+1}, ..., y_{2m} = 0$, giving

$$
\begin{aligned}
D &= \left(1 - \frac{1}{2}f\right)^{2m} \times \left(\frac{1}{2}f\right)^{-2m} && (13)\\
&= \left(\frac{2-f}{f}\right)^{2m} && (14)
\end{aligned}
$$

Therefore,

$$
\varepsilon = 2m \log\left(\frac{2-f}{f}\right) \tag{15}
$$

□

References

[1] Úlfar Erlingsson, Vasyl Pihur, and Aleksandra Korolova. Rappor: Randomized aggregatable privacy-preserving ordinal response. In *Proceedings of the 21st ACM Conference on Computer and Communications Security*, Scottsdale, Arizona, 2014.